

# The Math of Closing: Proration Rules and Methods

Although no one expects you to run out and get your CPA certification, it's important to understand how prorations are computed at closing. Just to confuse things, some areas do it one way, others do it another. Following are some general methods that we hope clarify the murky waters.

To prorate means to divide or distribute a sum of money proportionately. Prorations are used to divvy up expenses on the property at closing. Some items are paid by the seller and some by the buyer. The Closing Disclosure (CD) lists all prepaid expenses that the seller has already paid, generally as a credit to the seller and a debit to the buyer.

It's important to develop a solid understanding of calculating prorations. To ensure that the math is done correctly, you must ask yourself the following questions:

1. What is being prorated?
2. Is the item an accrued or a prepaid item?
3. Is it based on the statutory year(360 days) or the calendar year(365 days)?

## Prorated Expenses

Prorated items will be either accrued or prepaid. Both are shown on the closing statement.

**Accrued** expenses go in the seller's debit column. They are expenses that the seller owes at the day of closing but that the buyer will eventually pay once the buyer owns the property. The seller will give the buyer a credit for these items at closing. Typical prorated accrued items include:

- Unpaid real estate taxes
- Rent the seller collected from the tenant for a period of time in which the buyer will own the property
- Interest on the seller's mortgage that the buyer assumed (if applicable)
- Utilities billed and not paid in advance

**Prepaid** items are items the seller has paid, but that the buyer needs to credit the seller for the amount pertaining to the time period that the buyer will own the property. Prepaid expenses are divided proportionately between the buyer and the seller at closing and go in the seller's credit column and the buyer's debit column. Prepays include:

- Prepaid taxes
- Rent paid by the seller under a lease assigned to the buyer
- Utilities billed and paid in advance

Here's an example of prorated prepaids:

The seller prepaid \$565 for a security system, which the buyer intends to take over. Expenses are prorated through August 2, the day of closing. The proration will be calculated on a 360-day basis (the buyer owes the seller for the period between August 2 and December 31).

- $\$565 \div 360 = \$1.5694444$ , or \$1.569 per day
- 28 days in August plus 30 days each in September through December = 148 days
- $148 \times \$1.569 = \$232.212$ , or \$232.21 (debit to the buyer).
- Settlement statement will show a credit to the seller and debit to the buyer of \$232.21 to refund the seller for the portion of time that's been paid but during which the buyer will live in the property.

## Who pays the costs on the day of closing for property taxes, utilities, etc.?

Typically the seller pays the costs on the day of closing. Some local practices provide that beginning with the day of closing, all expenses are the buyer's, and all income (e.g., from a current renter) goes to the buyer. When you're taking your licensing exam, any questions about proration will tell you who owns the property on the day of closing so you can calculate properly.

## Calculating Prorations

These can be calculated one of two ways.

- **Using a statutory year:** Sometimes these items are prorated based on a 360-day statutory year (aka "banker's" year): 30 days x 12 months.
- **Using a calendar year:** The other method is to prorate based on a 365-day calendar year. These calculations use the exact number of days in each month of the year.

A note about insurance: amounts prepaid by the seller are likely to be prorated and refunded to the seller by the insurance company, rather than the buyer. Because insurance rates are based on a variety of factors like credit ratings, insurance claims, etc., the buyer will likely purchase a new policy.

### Calculating Assessed and Accrued Property Taxes That Aren't Yet Due

Normally, these are prorated to the day of closing. The seller will receive a debit and the buyer a credit for the amount owed as of the day of closing. Again, the calculations will include who actually "owns" the day of closing.

#### Example

Here's a sample item from our seller, Karen, and buyer, Vince.

The real estate taxes assessed on the property are \$785 for the year. The taxes aren't due until the following January 1. Closing is held on May 31. Here is how that would look on the closing statement:

Accrued period of taxes owed by Karen at closing using the statutory year:

- $\$785 \div 12 = \$65.42$  per month
- $\$65.42 \times 5$  full months (January 1-May 31) = \$327.10

Therefore, \$327.10 is the amount Karen owes for taxes accrued and not yet paid, and this amount will appear as a debit to Karen and a credit to Vince at closing.

If, instead of closing on May 31, closing were on June 15, this is a mid-month closing, and the partial month would be prorated by dividing the monthly amounts by 30 and then multiplying by the appropriate number of days. Since seller Karen will own the day of closing, the calculation is:

- $\$65.42 \div 30 = \$2.18$
- $\$2.18 \times 15 = \$32.70$

$\$327.10$  (January 1-May 31) +  $\$32.70$  (June 1-June 15) = \$359.80

### Calculating Prepaid Property Taxes

In many cases, the seller has already paid the property taxes. That means the buyer needs to reimburse the seller for the taxes that cover the time the buyer will own the property that year.

#### Example

A seller paid taxes of \$3,000 for the entire year on January 1, then sold the property, closing on June 1. Who owes what to whom?

The seller paid the full year's taxes but used the property for only five months (January through May). The buyer needs to pay the seller for the taxes already paid for the months the seller won't own the property during the year (June through December). Assume a statutory year.

- Divide the amount paid (\$3,000) by 12 for the monthly cost ( $\$3,000 \div 12$  months = \$250 per month).
- Determine how many months the seller used the property ( $\$250$  per month x 5 months [the time the seller owned the property] = \$1,250), the amount of taxes that the seller paid from January through May.
- Take the monthly cost and multiply by the number of months the buyer will be using the property ( $\$250$  per month x 7 months [the new buyer's time in the house] = \$1,750). This is the amount of the taxes the buyer will pay because he bought the house on June 1.

The seller had already paid the taxes for the whole year, so the buyer owes the seller \$1,750 for the taxes the buyer is using but didn't pay. The buyer gets a debit (\$1,750) and the seller gets a credit (\$1,750).

Of course, not all closings fall neatly on the last day of the month. For closings that occur in the middle of the month, you need to calculate a daily rate for taxes instead of a monthly rate.

#### Example

A seller paid taxes of \$2,920 for the entire year on January 1, then sold the property, closing on April 10. Assume a calendar-year method (non-leap year) and that the seller owns the day of closing. The buyer needs to pay the seller for the taxes already paid for the days the seller won't own the property during the year. Who owes what to whom?

The seller paid the full year's taxes but used the property for only three months and 10 days, for a total of 100 days.

- Divide the amount paid (\$2,920) by 365 for the daily cost ( $\$2,920 \div 365 \text{ days} = \$8 \text{ per day}$ ).
- Determine how many days the seller used the property, which we already did: 100 days. That means the buyer will own the property for 265 days.
- Calculate the taxes that are the buyer's responsibility:  $265 \times \$8 = \$2,120$ . That's the amount the buyer will need to reimburse the seller at closing because the seller has already paid the annual taxes.

Because the buyer owes the seller \$2,120 for the taxes the buyer should pay, the buyer gets a debit (\$2,120) and the seller gets a credit (\$2,120).

### **Prorating Rent**

The seller typically receives rent for the day of closing (but see above: sometimes the buyer receives all income beginning with the day of closing).